THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 287

THE HATCH JACK

THE STUCK-DOOR DOCTRINE

jaws of life not optional

the jam lives in the wall — equalize first, spread second

the grep found no stuck-hatch answer in the whole file — now it does

THE SPREADER KIT · THE WALL MAP · THE BUILT-IN DOOR

THE STUCK DOOR IS A WHEN, NOT AN IF

Rescue tooling — v1.0 in force

GP-IM-287

Revision v1.0 — 24 August 2026

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VETTED SITTING 425 — PRINCIPLE PASS · DOOR DETAIL OWED

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 287

PHASE 287: THE HATCH JACK — THE STUCK-DOOR DOCTRINE, THE MANDATORY SPREADER KIT — STATUS: CURRENT — MECHANICAL PRINCIPLE PASS / AUTHOR-DESIGN DETAIL + PRESSURE-EQUALISATION IMPLEMENTATION (CO-ENGINEER FRESH AUDIT, SITTING 425). The audit's grep found no stuck-hatch answer in the whole file — no pry, no crowbar, no jammed hatch across hatches, pod doors and Phase 19's blast gates. His ruling, verbatim: 'jaws of life not optional, bulkhead doors are driven from inside the walls, any jam is there, if its a power failure yes, but leave that, i will design the doors to have a built in...' The ruling as law: (1) the spreader is NOT optional — the jaws-of-life class mechanical spreader is mandatory ship's kit, carried, not wished for; (2) the jam lives in the wall — bulkhead doors are wall-driven, so the machinery jam is addressed at the wall; (3) the doors get a built-in answer by his design. The audit's arithmetic miss is owned and corrected in the open: the pitch claimed '0.1 bar across a square metre carries ten tonnes' — the correction seated with it (0.1 bar on a square metre is 1,000 kg-force — the miss owned, the number fixed). The stuck door is a when, not an if; the kit rides before the jam.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-287, v1.0 — 24 August 2026. The two hundred and eighty-eighth manual of the program — 288 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the jack law as seated (the gap call, his ruling verbatim, the ruling as law, the grade with the arithmetic miss owned and corrected — entire) — §2 the physics, graded — §3 the system at the door — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 287: **The Hatch Jack — the stuck-door doctrine: jaws-of-life class mechanical spreader carried as MANDATORY kit (screw-driven, no hydraulics, no power — works**

exactly when power has failed); bulkhead doors are wall-driven, so the jam lives in the wall — the spreader attacks the frame gap, the built-in emergency opening system (RESERVED TO THE AUTHOR) attacks the drive; power failure affirmed as a jam case. Audit law-candidate awaiting his word: equalization bleed at every hatch — crack the pressure, never the door. *[RESERVATION CLOSED 18 AUGUST 2026 — he spoke it (Amendment 2 in the body): the drive detaches onto a transverse slide rail clear of the runner channel, both leaves slide one way past a jammed centre lock, twin mechanical winders cover the dead-power case; PRE-VETTED, engineering dims remain.]*'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

SEATED 15 August 2026 — the gap call: the audit's grep found no stuck-hatch answer in the whole file (no pry, no crowbar, no jammed hatch, across hatches, pod doors, and Phase 19's blast gates). The ruling and the door design: the author, from the road.

AUTHOR (verbatim — the ruling, typos preserved): "jaws of life not optional, bulkhead doors are driven from inside the walls, any jam is there, if its a power failure yes, but leave that, i will design the doors to have a built in emergency opening system"

THE RULING AS LAW: (1) THE SPREADER IS NOT OPTIONAL — the jaws-of-life class mechanical spreader is mandatory ship's kit, carried, not wished for. (2) WHERE THE JAM LIVES — bulkhead doors are driven from inside the walls; the machinery is in the wall, so any jam is there: the spreader's target is the door-frame gap, the mechanism's target is the wall. (3) POWER FAILURE — affirmed as a jam case. (4) RESERVED TO THE AUTHOR — the doors will carry a built-in emergency opening system of HIS design; the audit holds off per the standing pattern (the Phase 282 precedent: his mechanisms are his). *[AUDIT LAW-CANDIDATE, AWAITING HIS WORD: the equalization bleed — crack the pressure, never the door; a bleed path at every hatch so a pressure-locked door is unlocked by a valve, not a hero.]*

AUDIT GRADE — AFFIRMED, WITH AN ARITHMETIC MISS OWNED AND CORRECTED: the pitch claimed '0.1 bar across a square metre carries ten tonnes' — WRONG BY TEN: 0.1 bar over 1 m^2 is 10 kN ≈ ONE tonne-force; ten tonnes is the full 1 bar over 1 m^2. The conclusion stands at either number — no crewman muscles a pressure-locked hatch — but the file corrects the figure here per doctrine. THE SPREADER PHYSICS: screw-driven mechanical advantage (no hydraulics to leak, no power to fail — the power-failure case is exactly when it must work), ratchet-walking a warped frame open a click at a time; jaws of life grown into ship's furniture. THE TWO-TOOL DOCTRINE HIS RULING CREATES: the jam lives in the wall (his domain — the built-in emergency system) and the gap lives at the frame (the spreader's domain) — together they cover drive-failure AND frame-warp, and the bleed (if adopted) covers pressure-lock. Bench: the author's emergency opening system (owed); spreader force rating and jaw travel; bleed-valve ruling.

AMENDMENT — 15 August 2026 — EQUALIZATION RULED: PRESSURE MATCHING, NOT BLEEDING. The author, verbatim: "where possible at least in hallways and the bridge, non pod areas, i will look at built in pressure equalization countermeasures, pressure matching, bleeding may not be appropriate or possible in many cases" — SEATED AS LAW (body and index both): built-in pressure-equalization countermeasures WHERE POSSIBLE, at least in hallways and the bridge — the non-pod areas; the mechanism is PRESSURE MATCHING, and bleeding is ruled inappropriate or impossible in many cases. The audit's bleed law-candidate is adopted only as corrected by this ruling; the countermeasure design itself is the author's ("i will look at" — his to draw).

AUDIT GRADE — AFFIRMED, AND THE CORRECTION IS RIGHT FOR TWO REASONS THE AUDIT'S CANDIDATE DID NOT PRICE: (1) BLEEDING SPENDS THE SHIP'S AIR — a bleed valve answers a pressure lock by throwing gas overboard or into the far volume; in the worst case that lock matters most (a breached, evacuated compartment on the other side) there is nothing to bleed TO — you would be venting your own hallway into the hole. (2) MATCHING MOVES AIR INSTEAD OF LOSING IT — transfer between compartments (pump/valve paths built into the wall-run machinery the doors already use) equalizes both sides with the gas staying inside the ship; the door unlocks and the air is still yours. SCOPE NOTE SEATED AS RULED: hallways and the bridge first, non-pod areas — the pods keep their own door doctrine; 'where possible' is the honest boundary, his words. Bench: his matching-countermeasure design (owed); which compartments rate the fit-out.

AMENDMENT 2 — 18 August 2026 — THE RESERVED MECHANISM SPOKEN: THE BUILT-IN EMERGENCY OPENING SYSTEM, DICTATED AND PRE-VETTED. The reservation he laid down in the ruling ("leave that, i will design the doors to have a built in emergency opening system") is collected. His order and the design entire, verbatim, typos preserved: "287 yes i had forgotten that i will invent that now, kimi can add it to 287, pre-vetted — as these relate to bulkhead door with single pods pressure variance being of low danger component is either direction of vacuum and pressure. a per section under floor plumbing array per section to pressure tanks and the outer vacuum will equalize room pressures. the heavy duty mechanical drives of the bulkhead doors will be on an internal slide rail horizontally/perpendicular to the direct it operates. in an emergency, the drive mechanism will dethatch from the door inside the wall and will mechanically move sideways away from the door runner channel. Thus allowing the doors to be forced open. wherein the centre-lock is engaged and jammed both doors can simply be slid in one direction and access gained via the open end. where a power outage has occurred or control lines damaged. the built in mechanical winder in the door face will release the drive lock grips and the mechanical winder in the alternate room where the door slides between will have a similar wider to then move the drive mech sideways clear of the track. i think i got it all"

THE MECHANISM AS LAW (the co-engineer's vetting, seated): (1) PRESSURE DIRECTION NOT ASSUMED — bulkhead doors split separately controlled pressure sections; the differential can run either direction, or to vacuum on one side. (2) EQUALIZE WHEN POSSIBLE — a per-section under-floor plumbing array to pressure tanks brings adjacent rooms toward match for NORMAL operation (the Amendment-1 doctrine rides: pressure matching, not bleeding). (3) THE DRIVE LEAVES THE LOAD PATH — the heavy-duty drive rides an internal slide rail perpendicular to the door's travel; in emergency it detaches from the door inside the wall and moves mechanically sideways, clear of the runner channel — the door no longer fights its own failed drive. (4) CENTRE-LOCK JAM BYPASSED — with the centre lock engaged and jammed, both leaves slide together in one direction and access is gained via the open end; the centre lock is removed from the emergency-opening requirement. (5) FULLY MECHANICAL RECOVERY CHAIN — the door-face winder releases the drive-lock grips; the matching winder in the alternate room (where the door slides between) moves the drive sideways clear of the track; the chain holds with power out, control lines damaged, actuator seized, centre lock jammed.

AUDIT GRADE (co-engineer Gemini 1, concurred by this audit): PRE-VETTED MECHANICAL PRINCIPLE — the standing flag 'invent the emergency-opening mechanism' (reserved to his hand since 15 August) is CLOSED BY HIS DESIGN. Ordinary engineering remains, as the co-engineer listed: size the transverse actuator slide for the door/pressure loads; the drive cannot re-engage accidentally; positive mechanical indication it has cleared the runner; manual force calculated under the maximum specified pressure differential; the emergency winders accessible and not trappable by the same failure; the complete sequence physically tested with BOTH pressure directions. And the separation is law: equalization is a normal-operation aid, NOT an emergency-access requirement — equalize when possible for normal operation; when you cannot, or the drive has failed, mechanically remove the drive and bypass it. The independent failure path is the point.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE GREP WAS THE GAP CALL: the audit searched the whole file for the stuck-hatch answer and found none — no pry, no crowbar, no jammed hatch across every door the ship carries. The phase exists because the absence was measured, not felt.

THE SPREADER IS NOT OPTIONAL: the jaws-of-life class mechanical spreader is mandatory ship's kit — carried, not wished for. The doctrine is rescue-tool logic: the stuck door is a when, and the kit rides before the jam.

WHERE THE JAM LIVES: bulkhead doors are driven from inside the walls — the machinery jam is in the wall, so the access doctrine addresses the wall; 'if its a power failure yes, but leave that' — power restoration is the first pry.

THE BUILT-IN ANSWER IS HIS DESIGN DEBT: 'i will design the doors to have a built in...' — the author-design detail is the vet's named remaining work, and the phase carries it as owed.

THE ARITHMETIC MISS IS OWNED IN THE OPEN: the pitch claimed 0.1 bar across a square metre carries ten tonnes — the true figure is one tonne (0.1 bar = 10 kPa; on 1 m² that is 10 kN ≈ 1,000 kg-force). The correction is seated beside the claim — the file's own miss, fixed in the record.

PRESSURE-EQUALISATION IS THE HONEST FIRST MOVE: a pressure-locked door is not a jam — equalize before you spread; the vet's implementation flag prices the sequence.

§3 — THE SYSTEM AT THE DOOR

- **The gap:** no stuck-hatch answer existed — the grep proved the absence.
- **The kit:** the jaws-of-life spreader — mandatory, carried, not optional.
- **The jam address:** the wall — bulkhead doors are wall-driven; the machinery lives there.
- **The built-in:** the doors' own answer — his design, owed and named.
- **The correction:** 0.1 bar on a square metre is one tonne, not ten — the miss owned.
- **The sequence:** equalize first, spread second — pressure lock is not a jam.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phase 19's blast gates were in the grep's scope — every door class the ship carries now has its stuck answer.
- 285's rescue doctrine is the sibling: the stuck hatch and the strange hull are the same family — the crew gets through or the crew gets out.
- The arithmetic-miss pattern is the audit's honesty habit: the correction seated beside the claim, in the open, no deletion.
- 282's tear-down method is the cousin: the failure modes found first, the design built on the catches.

§5 — THE VET RECORD

THE CO-ENGINEER'S FRESH AUDIT (PHASES 278-290), SEATED IN SITTING 425 (18 August 2026). The grade, verbatim from the audit: '287 → mechanical principle PASS / author-design detail + pressure-equalisation implementation.' The remaining work named by the verdict itself: his door detail and the pressure-equalisation implementation.

§6 — BUILD SEQUENCE

- 1. Stock the spreader: jaws-of-life class, mechanical, carried per deck — the manifest entry.
- 2. Map the walls: every bulkhead door's drive machinery located and plated — the jam address chart.
- 3. Write the sequence: power check → equalization → spread — the drill card on the kit.

- 4. Build his door detail: the built-in answer per his design — the vet's named work.
- 5. Drill the stuck door: timed, in the dark, suited — the rescue cadence.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 19 (the blast gates):** the door class in the grep's scope — now covered.
- **Phase 285 (the rescue chutes):** the sibling — through or out, the rescue doctrine.
- **Phase 282 (the survival sphere):** the method cousin — the catches as the design.
- **Phase 286 (the leak hound):** the pressure cousin — equalization and drift share the air.

§8 — DO-NOT-BUILD

- Do NOT wish for the kit — the spreader is carried, mandatory, per deck.
- Do NOT spread a pressure-locked door — equalize first; a pressure lock is not a jam.
- Do NOT hunt the jam in the door — the jam lives in the wall; the wall is the address.
- Do NOT leave the built-in undesigned — his door detail is the named debt.
- Do NOT quote the ten tonnes — the correction is seated: 0.1 bar on a square metre is one.

§9 — OWED

OWED: the vet's pair, whole — the author-design door detail (the built-in answer, drawn and benched) and the pressure-equalisation implementation (the equalize-first hardware and drill). Plus the spreader spec and the per-deck stowage map.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-287, v1.0 — CURRENT — MECHANICAL PRINCIPLE PASS / AUTHOR-DESIGN DETAIL + PRESSURE-EQUALISATION IMPLEMENTATION (CO-ENGINEER FRESH AUDIT, SITTING 425), seated law, master v2.1 (numerical print order). The two hundred and eighty-eighth manual of the program — 288 of 290. Built 24 August 2026, eighth of the 15-manual 'do them all to 294' shift (the author's order, 24 August 2026, seated in sitting 624 — 'do them all to 294 i will give the last four to the vet before uploading, so its all genuinely vetted, the others we didnt alter, so im not bothered. i will do these'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622, 623 and 624): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up' / 'do them all to 294 i will give the last four to the vet before uploading, so its all genuinely vetted, the others we didnt alter, so im not bothered. i will do these'. The phase's own chain, all seated in the master: the contents line and the bodies (as seated); the gap call (the audit's grep — no stuck-hatch answer in the whole file); his ruling verbatim ('jaws of life not optional, bulkhead doors are driven from inside the walls...'); the ruling as law (the spreader mandatory; the jam in the wall; the built-in door answer his to design); the audit's arithmetic miss owned and corrected in the open (0.1 bar across a square metre is one tonne, not ten); the sitting-425 fresh-audit grade (mechanical principle PASS / author-design detail + pressure-equalisation implementation) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: his door detail and the equalization bench, or his word.

END OF MANUAL — PHASE 287